CryptoGuard AI

Demo Video Script & Submission Guide

NVIDIA NEMOTRON | NEBIUS AI CLOUD | POST-QUANTUM | NIST SP 800-90B | BEST APPS & AGENTS

Inventor: Swagata Banerjee | **Registration:** 23BCE0143 | **Institution:** VIT Vellore

Hackathon: Nebius × NVIDIA Global AI Hackathon 2026

Deadline: October 30, 2026 | **Video Duration:** 2 minutes

Quick Reference	
Live App	https://gitswagata1.github.io/cryptoguard-ai/
Auto Demo URL	https://gitswagata1.github.io/cryptoguard-ai/?demo
GitHub Repo	https://github.com/gitswagata1/cryptoguard-ai
Total Duration	2 minutes (120 seconds)
Sections	Intro (15s) → Scanner (35s) → Entropy (30s) → Chat (25s) → Close (15s)

0:00 1. Introduction

15 seconds

> **ON SCREEN:** Open the app at the Dashboard page (dark theme). Show the hero section with shield animation, tagline pills, and stats row. Browser at 110% zoom, fullscreen.

"Hi, I'm Swagata Banerjee from VIT Vellore. This is **CryptoGuard AI** — an AI-powered cryptographic security operations center built with **NVIDIA Nemotron** on **Nebius AI Cloud**.

It scans code for crypto vulnerabilities, analyzes entropy quality using NIST standards, and provides post-quantum migration guidance — all in one platform."

KEY THINGS TO SHOW

- Hero shield animation with orbital rings
- Feature pills: "Nemotron Ultra 253B", "NIST SP 800-90B", "Post-Quantum Migration"
- "How It Works" 3-step section visible as you scroll slightly

0:15 2. Code Scanner

35 seconds

> **ON SCREEN:** Click "Code Scanner" in sidebar → Click "Load Example" button → Pause to show the vulnerable Python code → Click "Scan for Vulnerabilities" → Wait for scan animation → Show results with findings and score ring.

"Let's scan some real vulnerable Python code. This file uses MD5 for passwords, hardcodes a secret key, uses Python's `random` module for tokens, and encrypts with DES in ECB mode."

(Click Scan — wait for the dual-ring spinner and progress bar to finish)

"Nemotron Ultra — a **253-billion parameter model** — found **8 vulnerabilities**. Two critical: the hardcoded key and MD5 hashing. Three high: insecure RNG, DES encryption, and ECB mode. Each finding comes with a **specific code fix**. The security score is 12 out of 100 — this code is dangerously broken."

KEY THINGS TO SHOW

- Scan-line animation sweeping across the code editor
- Score ring filling up to 12/100 in red
- Scroll through findings — hover over one to show the fix recommendation in cyan
- Severity badges: CRITICAL (red), HIGH (amber), MEDIUM (purple), LOW (green)

0:50 3. Entropy Lab

30 seconds

> **ON SCREEN:** Click "Entropy Lab" in sidebar → Keep 256 bytes in the input → Click "Generate & Analyze" → Show metrics filling in, heatmap coloring, gauge animating, AI analysis text appearing.

"Now the Entropy Lab. I'll generate 256 random bytes using the Web Crypto API and analyze their quality against **NIST SP 800-90B** standards."

(Click Generate & Analyze — show metrics animating)

"We compute four metrics locally — Shannon entropy, min-entropy via the MCV estimator, chi-squared, and serial correlation. The byte distribution heatmap shows uniformity across all 256 values.

Then **Nemotron Nano** — an 8-billion parameter model — gives a natural language quality assessment in under a second. We also compute extractable key bits using the **Leftover Hash Lemma** — that's provably secure randomness extraction."

KEY THINGS TO SHOW

- Quality gauge animating from 0 to the final percentage
- Heatmap cells filling with color (hover over a few to show byte values)
- Four metric cards with bar fills: Shannon (cyan), Min-Entropy (purple), Chi² (amber), Serial Corr (green)
- AI Analysis text box with natural language assessment

1:20 4. AI Chat

25 seconds

> **ON SCREEN:** Click "AI Chat" in sidebar → Click the suggestion pill "How do I make my code quantum-safe?" → Wait for the typing indicator and response → Scroll through the formatted markdown reply.

"Finally, the AI Chat. I'll ask: **how do I make my code quantum-safe?**"

(Click the suggestion button — show typing indicator, then response appearing)

"CryptoGuard recommends replacing ECDH with **CRYSTALS-Kyber** for key exchange, ECDSA with **Dilithium** for signatures — these are the new NIST post-quantum standards finalized in 2024. It gives a concrete migration checklist with library recommendations."

KEY THINGS TO SHOW

- Suggestion pills at the top of the chat
- Typing indicator animation (three bouncing dots)
- Formatted response with bold keywords in cyan, bullet points, code blocks
- AI avatar with gradient glow effect

1:45 5. Closing

15 seconds

> **ON SCREEN:** Click "Architecture" in sidebar → Briefly show the patent claims and prior art section → Toggle to light theme (click moon icon) → Switch back to Dashboard to show updated stats → End on Dashboard.

"What makes CryptoGuard unique — it's the **only tool** that combines AI code scanning, NIST entropy analysis, and post-quantum guidance in one platform.
It uses **dual-model routing** — Ultra for deep reasoning, Nano for fast feedback — both running on **Nebius AI Cloud**.
Seven patent claims. Ten vulnerability classes. One unified security operations center.
Thank you."

KEY THINGS TO SHOW

- Architecture pipeline with animated arrows
- 7 patent claims scrolling briefly
- Theme toggle — show light mode works too (quick toggle)
- Dashboard with updated stats (scans: 1, vulns: 8, entropy score, chats: 1)

RECORDING TIPS

- **Software:** Use OBS Studio (free) or Windows Game Bar (Win+G) to record screen + mic
- **Resolution:** 1920×1080 (1080p), 30fps minimum
- **Browser:** Chrome or Edge, fullscreen (F11), zoom 110% for readability
- **Audio:** Speak clearly and steadily. Record in a quiet room. Use a headset mic if available
- **Backup:** The `?demo` URL auto-walks through all features — use it as a practice run
- **Light theme:** Toggle briefly during the closing to show both themes work
- **Pace:** Don't rush. Pause 1-2 seconds after each click to let the animation play
- **Format:** Export as MP4, upload to YouTube (unlisted) or directly to Devpost

Section	Time	Duration	What to Show
1. Intro	0:00 – 0:15	15s	Dashboard hero, tagline, pills
2. Code Scanner	0:15 – 0:50	35s	Load example → Scan → 8 findings → Score 12/100
3. Entropy Lab	0:50 – 1:20	30s	Generate 256 bytes → Metrics → Heatmap → AI assessment
4. AI Chat	1:20 – 1:45	25s	Ask PQ question → Kyber/Dilithium recommendation
5. Closing	1:45 – 2:00	15s	Architecture → Theme toggle → Dashboard stats → End

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